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NORMALIZATION

Step 1: Identify the Entities and Their Attributes

From the relational schema, we identify the entities and their corresponding attributes:

1. **Programmer**
 - Programmer ID (Primary Key)
 - Date of Joining
 - Qualification
 - Specialization
 - Years of Experience
 - Programming Languages Known
2. **Software**
 - Software ID (Primary Key)
 - Date of Commencement
 - Date of Release
 - Status of Software
 - Development Duration
3. **Client**
 - Client ID (Primary Key)
 - Name
 - City
 - Street
 - House Number
 - Zip Code
 - Phone Number
4. **Project Leader**
 - Leader ID (Primary Key)
5. **Develop (Associative Table between Programmer and Software)**
 - Programmer ID (Foreign Key)
 - Software ID (Foreign Key)
6. **Commission (Associative Table between Software and Client)**

- Software ID (Foreign Key)
- Client ID (Foreign Key)
- 7. **Leads (Associative Table between Project Leader and Software)**
 - Leader ID (Foreign Key)
 - Software ID (Foreign Key)

Step 2: Normalize Each Table to 3NF

1. Programmer Table

This table is already in 3NF as it has:

- **Programmer ID** as the primary key.
- No partial dependencies (attributes depend fully on the primary key).
- No transitive dependencies.

Programmer ID	Date of Joining	Qualification	Specialization	Years of Experience	Programming Languages Known
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2. Software Table

This table is also in 3NF as:

- **Software ID** is the primary key.
- All attributes depend fully on the primary key.

Software ID	Date of Commencement	Date of Release	Status of Software	Development Duration
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3. Client Table

This table is normalized as well:

- **Client ID** is the primary key.
- All attributes depend fully on the Client ID.

Client ID	Name	City	Street	House Number	Zip Code	Phone Number
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4. Project Leader Table

This table is already simple and normalized:

- **Leader ID** as the primary key.

Leader ID

5. Develop Table (Associative Table)

This table represents a **many-to-many relationship** between **Programmer** and **Software**.

- **Primary Key:** Composite key (Programmer ID, Software ID).

Programmer ID	Software ID

6. Commission Table (Associative Table)

This table represents a **many-to-many relationship** between **Software** and **Client**.

- **Primary Key:** Composite key (Software ID, Client ID).

Software ID	Client ID

7. Leads Table (Associative Table)

This table represents a **one-to-many relationship** between **Project Leader** and **Software**.

- **Primary Key:** Composite key (Leader ID, Software ID).

Leader ID	Software ID
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Step 3: Final Normalized Tables

Table Name	Primary Key	Foreign Keys	Attributes
Programmer	Programmer ID	None	Date of Joining, Qualification, Specialization, Years of Experience, Programming Languages Known
Software	Software ID	None	Date of Commencement, Date of Release, Status of Software, Development Duration
Client	Client ID	None	Name, City, Street, House Number, Zip Code, Phone Number
Project Leader	Leader ID	None	None
Develop	(Programmer ID, Software ID)	Programmer ID, Software ID	None
Commission	(Software ID, Client ID)	Software ID, Client ID	None
Leads	(Leader ID, Software ID)	Leader ID, Software ID	None

Normalization of the Relational Schema

We will follow the same process to normalize the given relational schema step by step: **UNF** → **1NF** → **2NF** → **3NF**.

Unnormalized Form (UNF)

In UNF:

- Tables may contain **repeating groups** or **multivalued attributes**.
- No strict structure is applied.
- For the given schema, all data is stored in a **single table**.
- **Example:**

Programmer ID	Date of Joining	Qualification	Specialization	Years of Experience	Programming Languages Known	Software ID	Date of Commencement	Date of Release	Status of Software	Development Duration	Client ID	Client Name	City	Street	House Number	Zip Code	Phone Number	Leader ID
P001	2022-01-01	B.Tech	AI	3	Python, C++	S001	2022-02-01	2022-05-01	Completed	3 Months	C001	Alpha Corp	NY	Main St	123	10001	555-1234	L001
P001	2022-01-01	B.Tech	AI	3	Python, C++	S002	2022-03-01	2022-06-01	In Progress	4 Months	C002	Beta Inc.	SF	Market Rd	456	94105	555-5678	L002

Problem with UNF:

1. **Data redundancy:** Repeated data like Programmer details, Client details, and Software details.
2. **Update anomalies:** Updating the specialization of a programmer requires changes in multiple rows.
3. **Insertion anomalies:** Adding a new programmer without assigning them software is difficult.
4. **Deletion anomalies:** Deleting a software entry may remove all details about a programmer or client.

First Normal Form (1NF)

In 1NF:

- All columns contain **atomic values** (indivisible).
- **Repeating groups** are eliminated.

Changes Applied:

1. Break the table into multiple tables to ensure atomicity.
2. Separate **Programmer**, **Software**, **Client**, and **Leader** into individual tables.

Tables after 1NF:

1. Programmer Table

Programmer ID (PK)	Date of Joining	Qualification	Specialization	Years of Experience
P001	2022-01-01	B.Tech	AI	3

2. Programming_Languages Table

Programmer ID (FK)	Programming Language
P001	Python
P001	C++

3. Software Table

Software ID (PK)	Date of Commencement	Date of Release	Status of Software	Development Duration
S001	2022-02-01	2022-05-01	Completed	3 Months
S002	2022-03-01	2022-06-01	In Progress	4 Months

4. Client Table

Client ID (PK)	Client Name	City	Street	House Number	Zip Code	Phone Number
C001	Alpha Corp	NY	Main St	123	10001	555-1234
C002	Beta Inc	SF	Market Rd	456	94105	555-5678

5. Leader Table

Leader ID (PK)
L001
L002

6. Develop Table (Associates Programmer and Software)

Programmer ID (FK)	Software ID (FK)
P001	S001
P001	S002

7. Commission Table (Associates Software and Client)

Software ID (FK)	Client ID (FK)
S001	C001
S002	C002

8. Leads Table (Associates Leader and Software)

Leader ID (FK)	Software ID (FK)
L001	S001
L002	S002

Second Normal Form (2NF)

In 2NF:

- The table is in **1NF**.
- There are **no partial dependencies** (non-key attributes depend on part of a composite primary key).

Changes Applied:

1. Identify composite keys in associative tables.
2. Ensure all non-key attributes depend on the **whole** primary key.

New Tables:

1. Develop Table (No change):

| **Programmer ID** (FK) | **Software ID** (FK) |

2. Programming_Languages Table (No change):

| **Programmer ID** (FK) | **Programming Language** |

3. Commission Table (No change):

| **Software ID** (FK) | **Client ID** (FK) |

4. Leads Table (No change):

| **Leader ID** (FK) | **Software ID** (FK) |

Reason for 2NF:

1. Removed partial dependencies.
2. Ensured attributes depend on the **entire primary key** in associative tables.

Third Normal Form (3NF)

In 3NF:

- The table is in **2NF**.
- There are no **transitive dependencies** (non-key attributes depend on other non-key attributes).

Changes Applied:

1. Remove transitive dependencies (e.g., if a contact number depends on a programmer or client ID, move it to a new table).

New Tables:

1. Programmer Table

| **Programmer ID** (PK) | **Date of Joining** | **Qualification** | **Specialization** | **Years of Experience** |

2. Programming_Languages Table

| **Programmer ID (FK)** | **Programming Language** |

3. Software Table

| **Software ID (PK)** | **Date of Commencement** | **Date of Release** | **Status of Software** | **Development Duration** |

4. Client Table

| **Client ID (PK)** | **Client Name** | **City** | **Street** | **House Number** | **Zip Code** |

5. Contact_Info Table

| **Client ID (FK)** | **Phone Number** |

6. Leader Table

| **Leader ID (PK)** |

7. Develop Table

| **Programmer ID (FK)** | **Software ID (FK)** |

8. Commission Table

| **Software ID (FK)** | **Client ID (FK)** |

9. Leads Table

| **Leader ID (FK)** | **Software ID (FK)** |

Reason for 3NF:

1. **Transitive dependencies removed:** Phone numbers are moved to a **Contact_Info** table.
2. Ensures all non-key attributes depend only on the **primary key**.